

Data Science – Bivariate Analysis

1) Replace the NaN values with correct value. And justify why you have chosen the same.

```
sl_no          0
gender         0
ssc_p          0
ssc_b          0
hsc_p          0
hsc_b          0
hsc_s          0
degree_p       0
degree_t       0
workex         0
etest_p        0
specialisation 0
mba_p          0
status         0
salary         0
salary_zscore  215
dtype: int64
```

The `salary` column contained missing values corresponding to students who were not placed. Since these students did not receive any salary package, the missing values were replaced with **0**. This approach preserves the meaning of the data and avoids introducing misleading salary values that would result from using the mean or median.

2) How many of them are not placed?

```
Not Placed    67
```

3) Find the reason for non placement from the dataset?

	ssc_p	hsc_p	degree_p	etest_p	mba_p
status					
Not Placed	57.544030	58.395522	61.134179	69.587910	61.612836
Placed	71.721486	69.926554	68.740541	73.238041	62.579392

The data suggests that **lower SSC, HSC, and Degree percentages** are strongly associated with not being placed.

4)What kind of relation between salary and mba_p

	salary	mba_p
salary	1.000000	0.175013
mba_p	0.175013	1.000000

There is a weak positive correlation ($r = 0.175$) between MBA percentage and salary. Students with higher MBA percentages tend to receive slightly higher salaries, but the association is not strong. Other factors likely have a greater influence on salary than MBA percentage alone.

5)Which specialization is getting minimum salary?

```
specialisation
Mkt&Fin      200000.0
Mkt&HR       200000.0
Name: salary, dtype: float64
```

Both Marketing & Finance (Mkt&Fin) and Marketing & HR (Mkt&HR) have the same minimum salary of ₹200,000.

6)How many of them getting above 500000 salary?

```

specialisation  sl_no  gender  ssc_p  ssc_b  hsc_p  hsc_b  hsc_s  degree_p  degree_t  workex  etest_p  mba_p  status  salary
Mkt&Fin        120    M      68.8  Central  68.40  Central  Commerce  64.6  Comm&Mgmt  Yes    82.66  64.34  Placed  940000.0  1
               151    M      71.0  Central  58.66  Central  Science   58.0  Sci&Tech  Yes    56.00  61.30  Placed  690000.0  1
               178    F      73.0  Central  97.00  Others   Commerce  79.0  Comm&Mgmt  Yes    89.00  78.81  Placed  650000.0  1
Name: count, dtype: int64

```

Three students received salaries above ₹500,000, and all of them were from the Marketing & Finance specialization. The highest salary among them was ₹940,000.

7) Test the Analysis of Variance between etest_p and mba_p at significance level 5%. (Make decision using Hypothesis Testing)

```

F_onewayResult(statistic=np.float64(98.64487057324708), pvalue=np.float64(4.672547689133693e-21))

```

At the 5% significance level, there is sufficient evidence to conclude that the mean etest_p score differs significantly from the mean mba_p score. Therefore, a statistically significant difference exists between etest_p and mba_p.

8) Test the similarity between the degree_t (Sci&Tech) and specialisation (Mkt&HR) with respect to salary at significance level of 5%. (Make decision using Hypothesis Testing)

At the 5% significance level, there is sufficient evidence to conclude that the mean salary of Sci&Tech degree holders differs significantly from the mean salary of Mkt&HR specialization students.

9) Convert the normal distribution to standard normal distribution for salary column

```

      salary  salary_zscore
0  270000.0      -0.200292
1  200000.0      -0.951839
2  250000.0      -0.415019
4  425000.0       1.463849
7  252000.0      -0.393547

```

- The salary of ₹425,000 is above the average salary, as its Z-score is +1.464.

- The salaries ₹200,000, ₹250,000, ₹252,000, and ₹270,000 are below the average salary, since their Z-scores are negative.
- After standardization, the transformed variable (`salary_zscore`) has:
 - Mean ≈ 0
 - Standard Deviation ≈ 1

10)What is the probability Density Function of the salary range from 700000 to 900000?

```
Mean: 288655.4054054054
Expand Output 241958875
z1: 4.401410309664898
z2: 6.541421564220343
Porbability: 5.377445759013888e-06
```

The probability that a student's salary lies between ₹700,000 and ₹900,000 is approximately **0.00000538** (or **0.00054%**). This is an extremely small probability, indicating that salaries in this range are very rare in the dataset.

11)Test the similarity between the degree_t(Sci&Tech)with respect to etest_p and mba_p at significance level of 5%.(Make decision using Hypothesis

Testing)

`etest_p` and `mba_p` are **not similar** for Sci&Tech students at the 5% significance level

12)Which parameter is highly correlated with salary?

`etest_p` (**Employability Test Percentage**) has the highest correlation with salary, with a correlation coefficient of **0.1783**.

13) plot any useful graph and explain it.

The graph indicates that **Marketing & Finance students receive higher average salaries than Marketing & HR students**. Therefore, among the two specializations, **Mkt&Fin appears to offer better salary outcomes** in this dataset. This suggests that specialization may have an influence on salary packages received during placements.

